

Exoplanet detection

Teaching with AI (TAI)

Mirte van der Eyden[1], Andrea López Incera[2]

[1] Institut für Fachdidaktik, Universität Innsbruck
mirte.van-der-eyden@uibk.ac.at

[2] Departament de didàctica de la matemàtica i les ciències experimentals, Universitat Autònoma de Barcelona
andrea.lopez.incera@uab.cat

universität innsbruck

UAB Universitat Autònoma de Barcelona

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next VIBENSHUS GYMNASIUM

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Subject: Physics

Activity duration: 150 min

Lesson outline:

1. Introduction to exoplanets and detection methods (5 min)
2. Initial research with an AI assistant, checking its answers against reliable sources (15 min)
3. Planning and building a model transit experiment (40 min)
4. Collecting and analysing light-curve data (75 min)
5. Calculating the size of your group's real exoplanet (15 min)

Learning goals:

- Explain how a transit light curve can provide information about an exoplanet.
- Design a fair experiment by changing one variable and controlling the others.
- Collect, plot, and interpret light-intensity data.
- Reflect on how scientists know what they know (Nature of Science): for example, the difference between an observation and an inference, and why scientific knowledge is reliable but can change.
- Use an AI research mentor critically, checking whether its suggestions make scientific sense.

Solution: Your role as a teacher. The heart of this lesson is that students **design the model experiment themselves**. Please do **not** show a finished set-up, give step-by-step building instructions, or tell students what to change and measure. Provide the materials and let each group work out, together with Emmy, how to build the model, which planet sizes to test and how to keep the comparison fair. Your role is that of a second mentor: ask questions rather than give answers (for example "What in your box plays the role of the star?" or "What else, apart from the size, could change the dip?"), keep an eye on safety (cutting knives), and help groups that are completely stuck by pointing them to the next question or hint. Different designs in different groups are welcome: comparing them in the *Peer Check* shows students that there is no single scientific method.

Files to share with the students: this worksheet (student version), `Emmy_prompt_exoplanet_lesson.txt`, `Star_image_for_phone.png`, `Exoplanet_transit_analysis_notebook.ipynb`, and the planet file of each group's exoplanet (`Planet1_55CancriE.pdf` to `Planet8_Trappist-1G.pdf`).
For teachers only: `Teacher_TransitGraphs_KEY.pdf` (the transit graphs of all eight planets with the transit depth marked) and `Teacher_Overview.pdf` (all data of the eight planets for the whole exoplanet unit). Never share the overview with students: it also contains the answers of the chemistry lesson.

Exoplanet detection

Experimenting with help of AI

How are exoplanets detected? How can scientists infer properties of a planet that is many light-years away? In this lesson, you will build a model of a distant solar system with a star and an orbiting planet. By measuring the star's light curve, you can detect the model exoplanet. By testing different model planet sizes, you will investigate how planet size affects the transit signal.

Initial ideas

Question 1: How do you think that exoplanets can be detected?

Solution: **Expected:** any ideas are fine here; this question activates prior knowledge. Typical answers: "with a big telescope", "taking a photo", "the star gets darker". Keep the answers in mind and come back to them after the initial research.

Constructing new ideas

Assignment 1: Initial research

Use an AI assistant to research the questions below. Ask specific questions, and ask it to explain ideas at your level. You should be able to explain your answers to your classmates.

Check your sources. An AI assistant can sound confident and still be wrong, and it sometimes invents facts or even sources that do not exist. Use its answers as a starting point, and check the important claims:

- Ask the AI where its information comes from, and open those sources yourself. Does the source exist, and does it really say this?
- Prefer sources from scientific organisations, such as space agencies, universities and research institutes.
- Look for the same claim in a second, independent source.

Question 2: What are the most common ways that exoplanets are detected?

Solution: **Expected:** the *transit method* (a planet passing in front of its star dims the starlight a little; most known exoplanets were found this way, for example by the Kepler and TESS space telescopes) and the *radial-velocity* or "wobble" method (the planet makes its star move slightly, which shifts the colour of the starlight). Other methods: direct imaging, gravitational microlensing, and astrometry or timing variations.

Question 3: What is a transit, and what information can we gain from it?

Solution: **Expected:** a transit is the moment when a planet passes between its star and us, blocking a small part of the starlight. From transits we learn the *size of the planet* compared with its star (from the transit depth), the *orbital period* (from the time between transits), and from that the size of the orbit. During a transit, starlight also passes through the planet's atmosphere, which tells us which gases it contains.

Question 4: What is a transit lightcurve? Sketch a graph of the typical shape of a transit lightcurve.

Solution: **Expected:** a graph of the brightness of the star against time. The sketch shows a flat line, a drop when the planet moves onto the star, a (nearly) flat or slightly rounded bottom while the planet is in front of the star, and a rise back to the original level when it leaves: a "bathtub" shape.

Question 5: How would the transit graph look for a bigger exoplanet?

Solution: **Expected:** the dip is deeper, because a bigger planet blocks more light. (The transit also lasts a little longer, because the planet needs more time to move fully onto and off the star.) Some students think a bigger planet only makes the dip longer; the experiment lets them test this.

Question 6: What is *transit depth?*

Solution: **Expected:** how much the brightness drops during the transit, as a fraction or percentage of the normal brightness: depth = (baseline − minimum)/baseline. Some students will already find that depth $\approx (R_{\text{planet}}/R_{\text{star}})^2$; that is fine, they can test this claim with their own experiment.

Question 7: Which reliable source did you use to check the AI's answers? Write down the source, and explain in one sentence why you trust it.

Solution: **Expected:** for example the exoplanet pages of NASA or ESA, the NASA Exoplanet Archive, a university or observatory website, or a textbook. Good reasons: written by experts or a scientific organisation, up to date, based on evidence and references, consistent with a second source.

Question 8: Compare the AI's answers with your source. Did they agree? Describe one thing that was different, missing, or that you could not check.

Solution: **Expected:** answers vary. Typical findings: the AI gives an outdated number of known exoplanets (it keeps growing), leaves out details, oversimplifies, or gives a link or reference that does not exist. Discuss briefly why checking matters, even when the AI was right.

Assignment 2: Model an exoplanet

The challenge

Goal: Build a model of an exoplanet transit and investigate how the size of the planet influences the transit light curve.
At the end of this class, you should deliver:

- One figure of a transit, showing light intensity as a function of time.
- One plot of your results, showing how the transit depth depends on the size of the planet.
- A five-sentence conclusion about your research.

Constructing new ideas

Start a chat with Emmy, your AI research mentor. Emmy is not an app: she is a prompt that you paste into a chatbot. Open the file `Emmy_prompt_exoplanet_lesson.txt` that your teacher shares with you, copy all of its text, and paste it as the first message in a new chat with the AI chatbot your school allows. Emmy will introduce herself and take it from there.
Emmy does not give you a recipe to follow. She helps you set up your own investigation by asking questions, one at a time. You can ask her questions in return, and you can ask her for a hint whenever you get stuck. Remember: the AI is a helper, not the final authority. You are responsible for checking whether your plan is safe, fair, and scientifically reasonable.

Real research is not a straight line. Scientists do not follow one fixed "scientific method". They move back and forth between different practices as their ideas develop. Emmy keeps a **research map** at the top of each message, so you can see where you are:

- **Question:** what exactly do you want to find out?
- **Model:** what do you think is going on, and what do you predict?
- **Design:** how will you test it fairly?
- **Measure:** collect your data carefully and honestly.
- **Make sense:** what do your data actually show?
- **Share:** what can you claim, based on which evidence?

Going back to an earlier step, for example after a surprising measurement, is part of good science. A few times, Emmy will also ask you to 🔍 **think like a scientist**: these questions are about how science itself works.

Solution: **Expected physics (Make sense):** the fraction of the star's light blocked during a transit equals the ratio of the projected areas, so the *transit depth* is

$$\text{depth} = \frac{\Delta F}{F} = \left(\frac{R_{\text{planet}}}{R_{\text{star}}}\right)^2 .$$

For a fixed star, depth $\propto R_{\text{planet}}^2$, so a graph of transit depth versus planet size squared should be an approximately straight line through the origin, with slope $1/R_{\text{star}}^2$. Larger model planets should therefore produce deeper transits.

Note for teachers (Nature of Science): Emmy is designed to let students discover this relationship from their own data, so the student version of this worksheet does not mention "size squared". During the session she asks two or three *Think like a scientist* questions, each linked to one aspect of the Nature of Science: science is empirical; observation differs from inference (astronomers almost never see an exoplanet, they infer it from a dip in starlight); scientific knowledge is tentative; science involves creativity; expectations shape what we see; science is social (peer review); and there is no single scientific method. The *Peer Check* and the last question of the lesson are good moments to discuss these aspects with the whole class.

Note for teachers (scaffolding the squared relationship): students rarely spot depth $\propto$ size^2 from a graph alone, so three steps guide them to it. (1) The notebook first plots transit depth against planet *size*; the points curve upwards. (2) The grid task *From width to area* (research log, after Measure) lets students discover that a circle twice as wide covers four times as many squares, so a planet twice as wide blocks four times as much light. (3) Section 6 of the notebook lets students test models themselves (`x = size`, `size**2`, `size**3`) and compare how close their points lie to a straight line through the origin. Emmy offers the same steps as hints, one at a time, when a group asks for help. If a group is stuck, point them to the next step rather than to the answer.

Get to work with the help of your AI research mentor Emmy. Below, you find a list of the available materials. After that, you can keep your research log.

Available Materials

- Light sensor on your phone. Recommended applications: phyphox (Android only), Physics Toolbox or Arduino Science Journal.
- Image of a star (`Star_image_for_phone.png`), displayed on a phone and used as the star light source
- Cardboard box
- Sticks
- Cutting knife
- Tape
- Modeling clay
- Compass
- Geo triangle
- Caliper

Use of AI for data analysis and programming: You may use AI to help with all programming steps, including organising your data, writing code, calculating transit depth, making plots, and improving the layout of your figures. You may also ask AI to explain error messages or suggest better ways to analyse your measurements. However, you remain responsible for checking the results: make sure the code uses the correct data, the axes and units are labelled clearly, and the conclusions match your measurements.

Analysing your data with Google Colab

To analyse the data from your light-sensor app on your computer, use the ready-made notebook that your teacher shares with you:

`Exoplanet_transit_analysis_notebook.ipynb`

1. In your light-sensor app, record a transit and **export the data as a CSV file** (in phyphox, use the three-dot menu, *Export Data*, and choose *CSV (Comma, decimal point)*; in Physics Toolbox, use the share/export button; in the Arduino Science Journal, export the recording as CSV; any other app that exports time and light as CSV works too). Send the file to your computer. You can upload phyphox's `.zip` file as it is.

2. Open the notebook in **Google Colab**: go to `colab.research.google.com`, choose *File* → *Upload notebook*, and select the notebook file (or use the link your teacher gives you). Then **run the cells from top to bottom**. When prompted, **upload your CSV**.

3. The notebook draws your **light curve**, helps you compute the **transit depth**, and, once you have tested several planet sizes, helps you plot how the **transit depth depends on planet size**. You can then **test which relationship fits your data best**.

Research Log

Use the fillable boxes below to keep your research log, one box for each practice on Emmy's research map. Your notes do not need to be perfect, but they should show how your thinking develops. If you go back to an earlier step, add to that box and write down why.

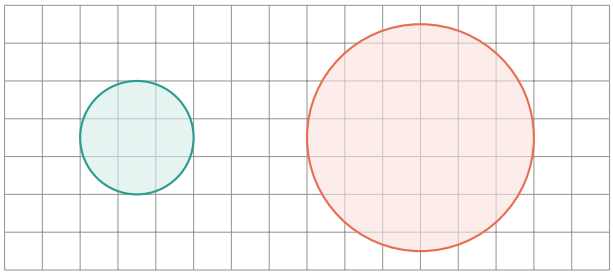
1. Question: *(What exactly do you want to find out? What will you change, and what will you look at to see the effect?)*

2. Model: *(What do you think happens to the light when the planet passes, and why? What does each part of your box represent, and in which ways is it different from a real star and planet?)*

3. Design: *(Which planet sizes will you test? How will you measure the dip? What will you keep the same between runs?)*

4. Measure: *(Record your measurements clearly, with units. Note odd runs, measurement problems, and anything you changed from your plan.)*

From width to area. *(Is your graph of transit depth against planet size a straight line? If not, this can help. A planet blocks the starlight with its whole disk, not only with its width.)*



A: diameter 1.5 cm B: diameter 3 cm

Count how many grid squares each circle covers (count a square if more than half of it is inside the circle). Circle B is twice as wide as circle A. How many times more squares does it cover? What does this tell you about how much light a planet that is twice as wide blocks?

5. Make sense: *(What pattern do you see in your data? Does it match what you expected in your model?)*

How did you calculate the transit depth? *(Explain your method in a few sentences, so that another group could repeat it.)*

6. Share: your conclusion in five sentences. *(1. Your answer to your research question. 2. The evidence from your data. 3. Why this evidence supports your answer. 4. How sure you are, and the biggest limitation. 5. What you would investigate next.)*

One suggestion for improving the experiment. *(What would you change if you had more time?)*

Think like a scientist. *(Choose one of the* Think like a scientist *questions Emmy asked you. Write down the question, your answer, and what it tells you about how scientists work.)*

Solution: Expected outcomes of the experiment (research log).

- *Question:* for example "How does the transit depth depend on the diameter of the model planet?" (change: planet size; look at: the dip in brightness).
- *Model:* a bigger planet blocks more light, so the dip is deeper. In the box, the phone screen is the star, the clay ball is the planet and the light sensor is the telescope; differences from reality: the star is flat and evenly lit, the "planet" is close to the star, and real dips are far smaller.
- *Design:* three or four planet sizes, moved across the centre of the star image at the same height, speed and distance from the screen, with the box closed. One run per size is enough.
- *Measure and Make sense:* the depth increases faster than the size; with a star image about 8 cm wide, a 2 cm planet gives roughly 6% and a 4 cm planet roughly 25%. Typical deviations: light leaking into the box, a planet that does not pass the centre of the star, a planet held far from the screen (blurred shadow), and automatic brightness changes of the phone. These are good material for the Make sense and Share discussion.
- *From width to area:* circle A covers about 7 squares, circle B about 28, so four times as many: a planet twice as wide blocks four times as much light.
- *Notebook section 6:* `x = size**2` gives the points closest to a straight line through the origin.

Checklist

Before you finish, make sure you have prepared:

- One labelled figure of a transit, showing light intensity as a function of time.
- One plot of your research results, showing how the transit depth depends on the size of the planet.

- A short explanation of how you calculated the transit depth (research log, box 5).
- A five-sentence conclusion about your research (research log, box 6).
- One suggestion for improving the experiment (research log, box 6).

Peer Check

Before scientists publish a result, other scientists check it. This is called *peer review*, and it is one of the reasons we can trust scientific knowledge.

Exchange results with another group and discuss:

- Are the axes, units, and labels clear on both graphs?
- Did both groups control the same variables during the experiment?
- Do the conclusions follow from the data, or are there alternative explanations?
- If your results differ, what could explain the difference?

Closing the challenge

Assignment 3: Calculating the size of your group's real exoplanet

Question 9: If you are given a transit light curve of a newly found exoplanet, how can you estimate its size? Use the transit-depth relationship in your explanation.

Solution: Model answer: rearranging depth $= (R_{\text{planet}}/R_{\text{star}})^2$ gives

$$R_{\text{planet}} = R_{\text{star}} \sqrt{\text{depth}}.$$

Measure the transit depth from the light curve, use the known (or modelled) star radius R_{star}, and compute R_{planet}. In the model experiment, R_{star} is the radius of the star-disk image on the phone, and depth = (baseline − minimum)/baseline.

Now that you have found the relationship between transit depth and planet size, you can use it to find out the size of a real planet.

From your model to your group's real exoplanet. Until now, you worked with your clay model planets. Now switch to the **real exoplanet your group is studying in the Exoplanet Files** (for example TRAPPIST-1 e or WASP-76 b). Open the **planet file** of your exoplanet that your teacher shares with you (for example `Planet3_Trappist-1E.pdf` for TRAPPIST-1 e). You need:

- the **single-transit graph** of your planet: read the transit depth from this graph;
- the **star radius** in the *Host star data*, in solar radii ($R_\odot$).

Write the transit depth and the planet radius in the empty fields of the datasheet in your planet file. Tip: 1 solar radius is about 109 Earth radii.

Question 10: Analyse the transit graph of your group's real exoplanet and calculate its radius. Show your calculation: the transit depth you read from the graph, the radius of the host star, and the radius of your exoplanet.

Solution: **Expected:** the transit depth of each planet is marked in `Teacher_TransitGraphs_KEY.pdf`, and all values (star radius, planet radius, transit depth) are in the *Physics & maths* table of each planet in `Teacher_Overview.pdf`. Worked example for TRAPPIST-1 e: depth 0.50% = 0.0050, so $R_{\text{planet}} = 0.119\,R_\odot \times \sqrt{0.0050} = 0.119 \times 0.0707\,R_\odot \approx 0.0084\,R_\odot \approx 0.0084 \times 109\,R_\oplus \approx 0.92\,R_\oplus$. Values within about 10 to 20% of the overview are a good result, since the depth is read from a noisy graph. A common mistake is using the depth in % instead of as a fraction (forgetting to divide by 100).

Question 11: How does the size of your exoplanet compare to the size of Earth?

Solution: **Expected:** depends on the planet (see `Teacher_Overview.pdf`). The eight planets range from about Earth-sized (TRAPPIST-1 e, TOI-700 d and TRAPPIST-1 g, about 0.9 to 1.1 $R_\oplus$) via super-Earths and sub-Neptunes (about 1.7 to 2.7 $R_\oplus$) to the gas giant WASP-76 b (about 21 $R_\oplus$, bigger than Jupiter).

Question 12: Your exoplanet orbits its own star, its *host star*. How does the size of this host star compare to the size of the Sun?

Solution: **Expected:** most host stars are red dwarfs, much smaller than the Sun: TRAPPIST-1 is only about 0.12 $R_\odot$ (hardly bigger than Jupiter), the others from about 0.2 to 0.9 $R_\odot$; only WASP-76 is bigger than the Sun (about 1.8 $R_\odot$). A good follow-up question: why are planets around small stars easier to detect with the transit method? (The same planet blocks a larger fraction of a small star, so the dip is deeper.)

What have we learned?

Question 13: What have you learned that you didn't know before?

Question 14: Astronomers have never seen most exoplanets directly. How is your model experiment similar to the way they find exoplanets, and how is it different? How sure can they be about the size of a planet they have never seen?

Solution: **Expected:** *Similar:* in both cases the planet is not seen but inferred from a dip in brightness, and the depth tells us the planet's size relative to its star. *Different:* real dips are tiny (from about 0.03% to 1.3% for the eight planets, against several percent in the model), real stars are glowing spheres that are darker at the edges, they are extremely far away, and real measurements are noisy, so many transits are combined. *How sure:* fairly sure, but the size depends on how well the star's radius is known, and it comes with an uncertainty; important results are checked with other telescopes and methods. This links to the Nature of Science ideas *observation versus inference* and *tentative but reliable knowledge*.

Question 15: How have you used AI in this lesson, and what did you learn from it?

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